23. The Concept of Abstraction

Abstraction, the Very Idea

Abstraction is the transition from a more specific (**low level**) description of some system to a more general (**high level**) description of that system . Abstraction always involves both a loss of information and a gain in generality. But abstractions are not arbitrary deletions of detail. Instead, they capture **"real patterns"** in lower-level phenomena.

For example, consider the following two systems made up of circles.



We can describe each system in terms of the relative location of colored **circles**. Or we can describe each as conforming to the shape of a **square**.



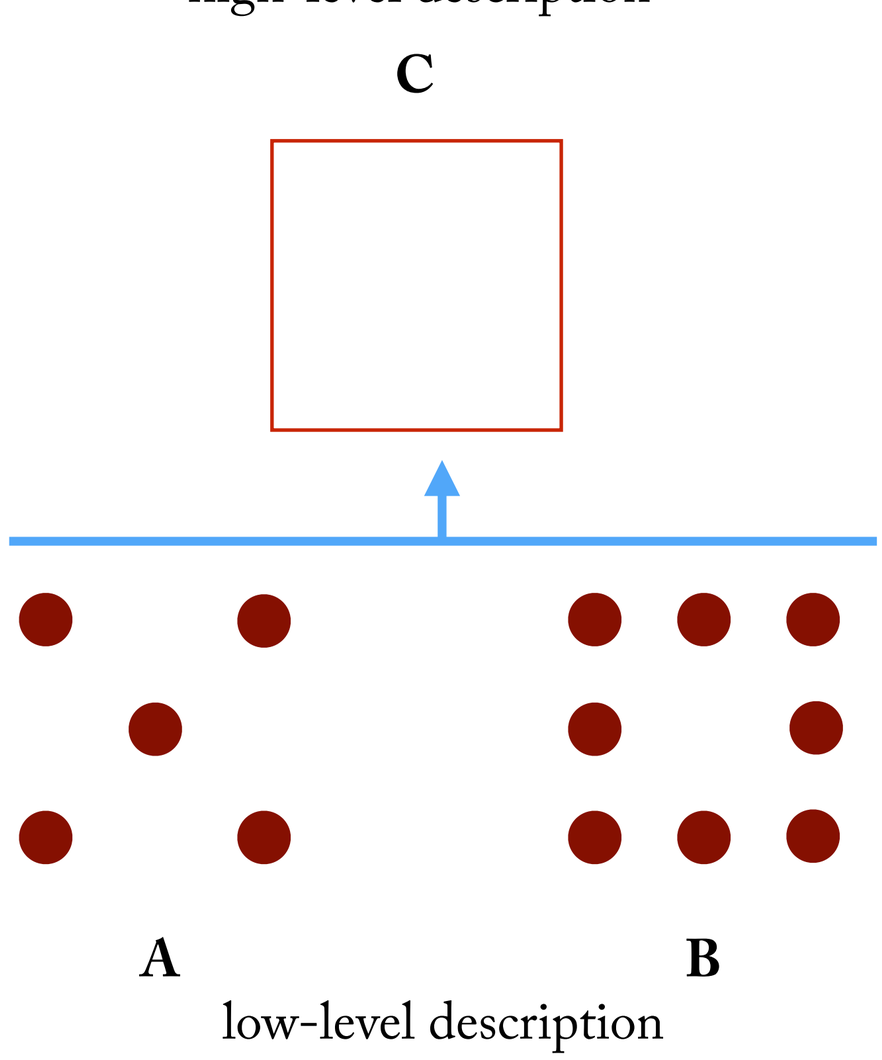
The relationship between the lower level and higher level descriptions here has many of the signature feature of abstraction in general.

Multiple realizability is the hallmark of abstraction. A description X is multiply realizable iff there is more than one (more) concrete description that can realize X. C is realized by both A and B.

Loss of detail. The high-level description C leaves out the composition and distribution of the circles that make up A and B.

Addition of deep structure. C explicitly introduces a level of structure, a "real pattern," which is present in both A and B, but explicit in neither.

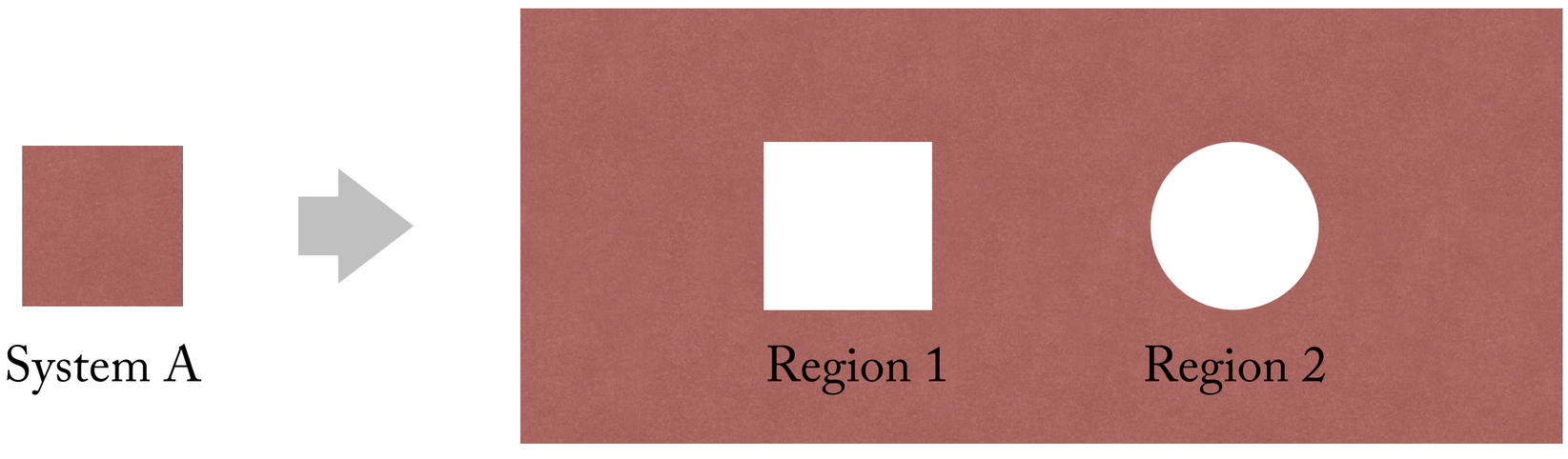
Autonomy. We can understand and reason about a system as an instance of C, without knowledge of whether it is A or B.



Abstraction as Explanation

Putnam's puzzle (from "Philosophy and our Mental Life," 1975):

> "Suppose we have a very simple physical system - a board in which there are two holes, a circle one inch in diameter and a square one inch high, and a cubical peg one-sixteenth of an inch less than one inch high. We have the following very simple fact to explain:
>
> *the peg passes through the square hole, and it does not pass through the round hole.*" (p. 295)



Consider two possible explanations of this fact.

> **Explanation 1:** microphysical deduction.
> System A is rigid lattice of atoms where atom a_1 is in position k_1 with features f_1, a_2 is in position k_2 with features f_2, … (and so on for every atom in A). It follows from the laws of quantum mechanics that, among all possible trajectories of A through space-time, A passes through R1 in some, but never passes through R2
>
> **Explanation 2:** geometrical deduction.
> A is a square with sides just under 1 inch across, and diagonal just under $\sqrt{2}$ inches. Since R1 is a square shape with sides 1 inch across, A passes through R1. Since R2 is 1 inch across at its widest point, and A is $\sqrt{2}$ inches across at its widest point, and $\sqrt{2}>1$, A does not pass through R2.
>
> **Verdict:** clearly E2 is the superior explanation; it's not clear that E1 is an explanation at all. But why? (p. 295)

In defense of the high-level explanation: **relevance.**

- "In this explanation certain *relevant structural features of the situation* are brought out."
- "*The ultimate constituents don't matter…only the higher level structure matters.*"
- "The explanation at the higher level brings out the relevant geometrical relationships. The lower level explanation conceals those laws. Also notice that the higher level explanation applies to a much more interesting class of systems."

In defense of the high-level explanation: **generality.**

- "The same explanation will go in any world (whatever the microstructure) in which those *higher level structural features* are present. In that sense *this explanation is autonomous*."
- "The higher level explanation is far more general, which is why it is *explanatory*."
- "An explanation is superior if it is more general."

The Moral: abstraction plays an essential role in any successful scientific explanation.

Functional Abstraction

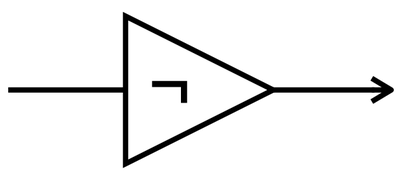
There are many different kinds of abstraction. We may describe a planet as an irregularly shaped solid, as a ball, or as point. We may describe a person as loose affiliation of particles, as a coordinated system of cells, as an animal, or as a rational agent.

In cognitive science and computational theory, **functional abstraction** dominates. This is a kind of abstraction that focuses in on a system's *function* or *purpose*, and abstracts away from details of how it achieves that function. Still there are many kinds of functional abstraction. We've already encountered several!

Functional Abstraction 1: Input-Output Abstraction

Logic gates themselves can be thought of as functional abstractions which make types of inputs and outputs relevant and let everything else fade away.

They abstract away from the internal description of a machine by specifying only inputs and outputs. And they abstract away from the external description by specifying inputs and outputs only as "0" and "1".



A NOT-gate.

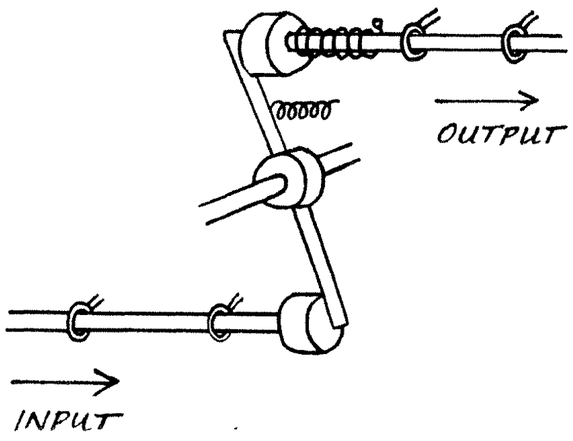
IN	OUT
0	1
1	1

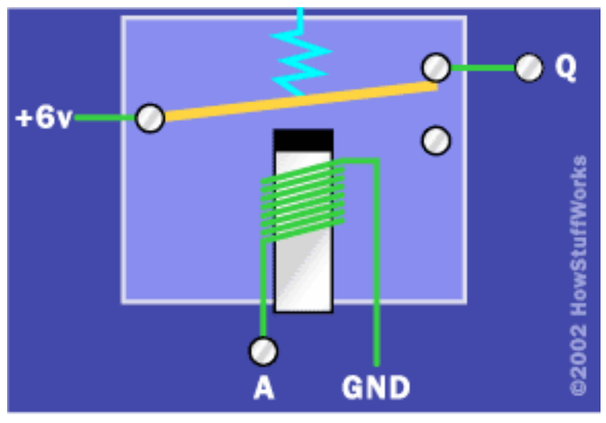
Abstraction over external mechanics.

IN	OUT
→	←
←	→

IN	OUT
⚡	⚡
⚡	⚡

Abstraction over internal mechanics.

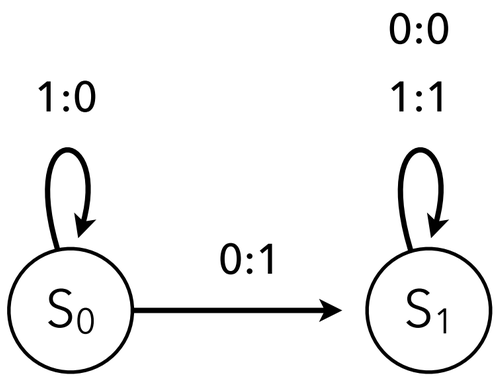




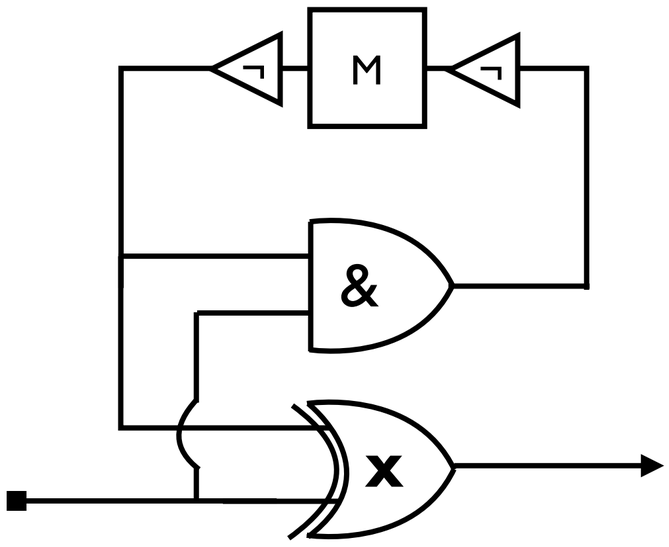
Physical implementations of a NOT-gate.

Functional Abstraction II: Piecewise Abstraction

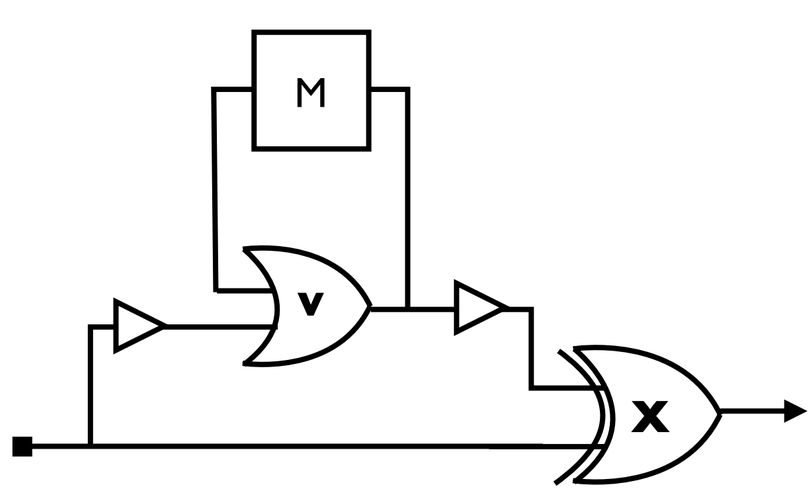
Functional Abstraction III: State Abstraction



+1 B FSM



+1 B SC



+1 B SC

Abstraction over internal machinery.

In state abstraction we find all of the central hall marks of abstraction itself.

Multiple realizability: Multiple SC's realize the same FSM.

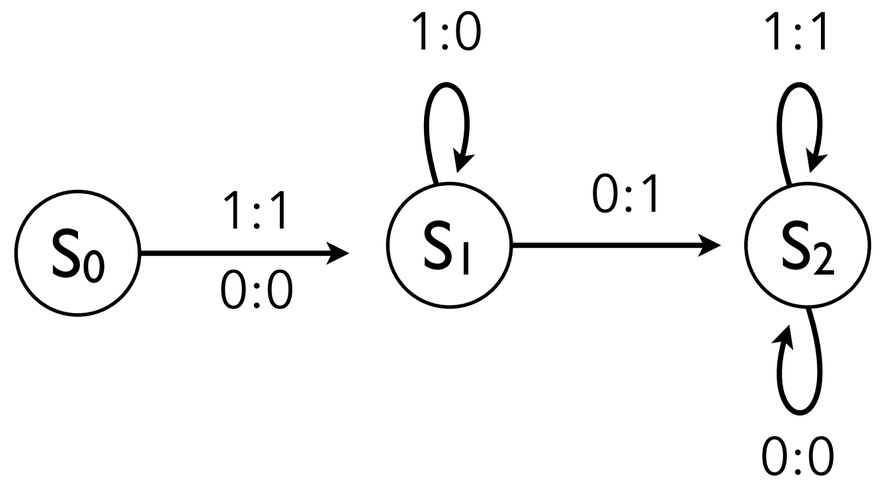
Loss of detail. The FSM leaves out details of circuitry.

Addition of deep structure. The FSM reveals the inner procedure that each SC is following.

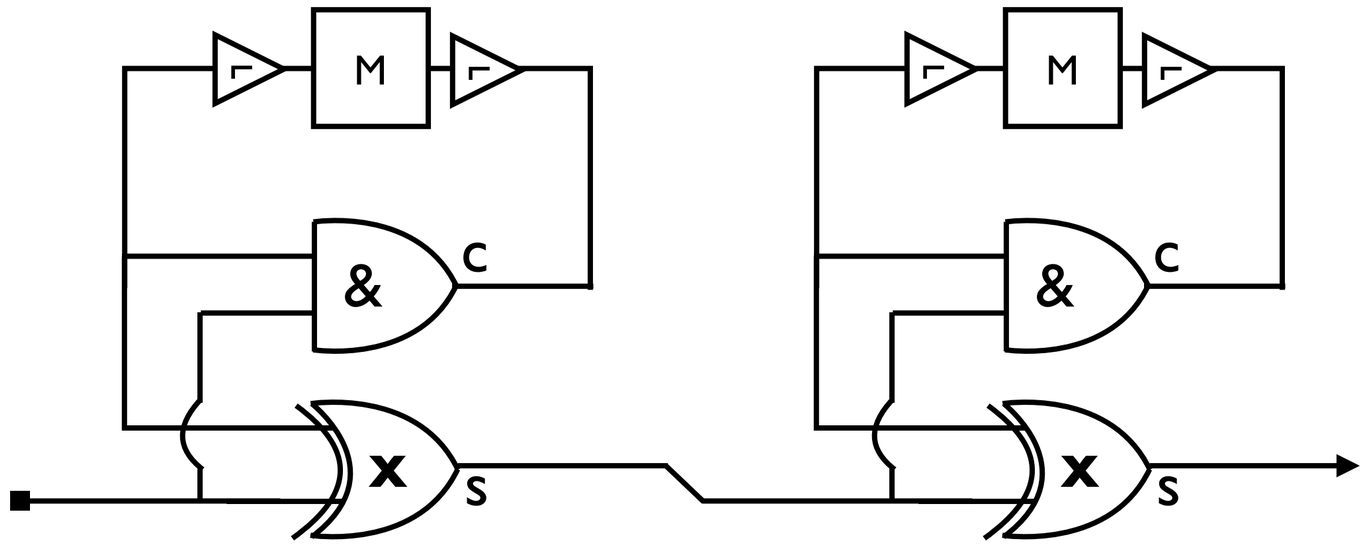
Autonomy. We can understand and reason about an FSM without understanding SC's, and visa versa.

State Machines and Sequential Circuits

FSMs describe pathways through overall configurations of the system. As the system changes, it flows through the state diagram. Each node represents a *temporal stage* in the machine's history.



SCs describe pathways of information. Information flows through the circuit as computation is carried out. Each nodes represents a *spatial* part of the machine.



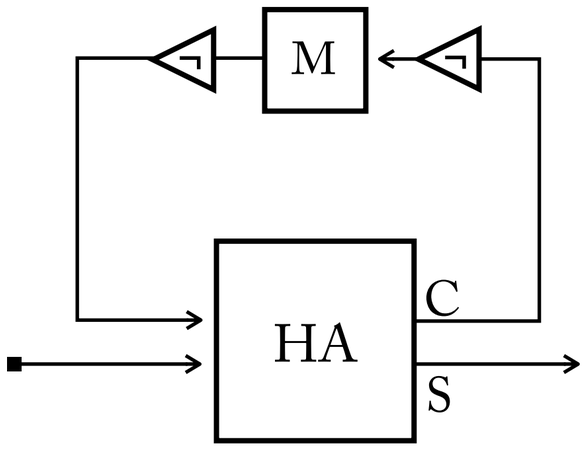
Both information flow diagram and state flow diagram describe systems, but in different ways.

We can think of a human as consisting of a complex circuit. Or we can think of them as implementing a complex algorithm and moving from state to state. Here we see the seeds of the contrast between neuroscience and psychology.

State abstraction marks an important kind of independence from the physical level. They are the first indication that computations can be carried out by neurons.

State Abstraction and the Mind-Body Problem

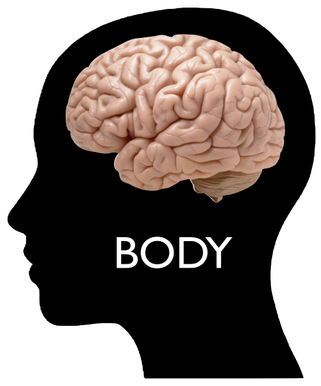
Consider now two very different descriptions of the same machine:

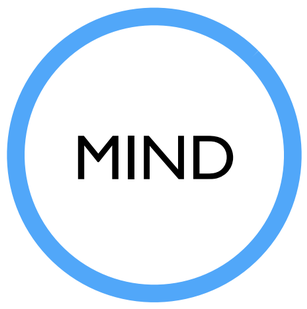


+1 B SC

STATE	IN	OUT	NEXT
S_0	0	1	A
	1	0	S_0
A	0	0	A
	1	1	A

+1 B State Table





Putnam (1967) "The Nature of Mental States" argued that state abstraction was the key to unlocking the mind-body problem. He thought the SC-State Table relationship was essentially equivalent to the body-mind relationship.

Since there is nothing mysterious about the way that SC's determine State Tables, there should be nothing mysterious about body-mind relationship. The implicit idea is that *mental states* are just *machine states*, for the right sort of human machine.

"I shall, in short, argue that pain is not a brain state, in the sense of a physical-chemical state of the brain (or even the whole nervous system), but another *kind* of state entirely. I propose the hypothesis that pain, or the state of being in pain, is a functional state of a whole organism." (Putnam 1967)

"Even though octopus and mammal are examples of parallel (rather than sequential) evolution, for example, virtually identical structures (physically speaking) have evolved in the eye of the octopus and in the eye of the mammal, notwithstanding the fact that this organ has evolved from different kinds of cells in the two cases. Thus it is at least possible that parallel evolution, all over the universe, might *always* lead to *one and the same* physical 'correlate' of pain… Thus any animal that we count as capable of these various states will at least seem to have a certain rough kind of functional organization."

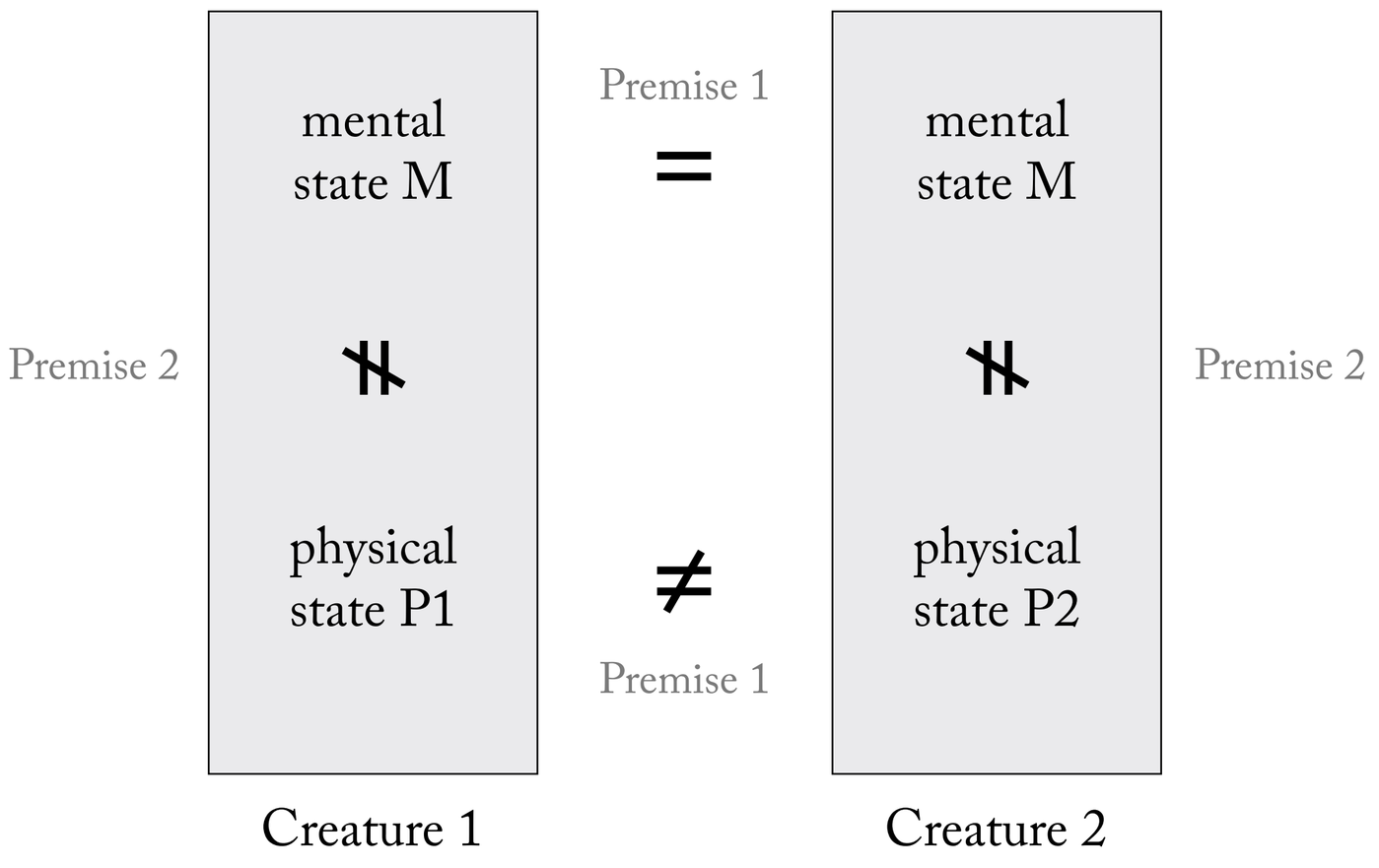
An Argument for Dualism?

1. In many cases, the same mental state can be physically realized in different ways by different creatures. Examples: beliefs, thoughts, perceptions, emotions.

2. Identity is transitive. Therefore: the mental states from Premise 1 cannot be identical with the physical states they are realized by.

3. According to physicalism, mental states are identical with physical states.

4. Therefore, physicalism is false.



"Anyone who wishes, then, to argue on this basis for the existence of the soul will have to be prepared to hug the souls of Turing machines to his philosophical bosom!" (Putnam 1960)